Rahul Pamadi

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Education

William & Mary **August 2021 – May 2025**
B.S. in Physics, Honors (GPA: 3.69, Major GPA: 3.85) *Williamsburg, VA*

Projects

Neural Vocal De-esser **2026**
Independent Project

- Built a synthetic paired-data training scheme for neural sibilance reduction, modeling harsh sibilance as a randomized dynamic equalizer gated by a sibilance detector.
- Designed and trained a Keras dilated temporal convolutional network which predicts time-frequency gain masks used to intelligently process vocal recordings for "Dancing Alone" by Emmett and the Dead Ends.

AI Synthesizer (Honors Capstone Project) **August 2024 – May 2025**
William & Mary Physics Department *Williamsburg, VA*

- Leveraged PyTorch and scikit-learn Python libraries to develop a convolutional neural network (CNN) which classified fast Fourier transforms of audio samples into a set of perceptual adjectives.
- Employed SPICE-based simulation and physical prototyping to develop and rigorously test circuits in a modular analog signal-processing chain, such as a preamplifier, equalizer, soft-clipper, tremolo, ring modulator, and phaser.

Experience

Scientist **July 2025 – Present (Full-time)**
NSWC: Dahlgren Division | King George, VA **May 2023 – August 2024 (Intern)**

- Developed and integrated a multi-purpose test program to completely automate the collection of radio-frequency spectral, waveform, power, and S-parameter data from electronic devices through Python control of an oscilloscope, spectrum analyzer, arbitrary waveform generator, and vector network analyzer.
- Test automation accelerated test timelines by 4x, saving projects a total of $120k.

Research Assistant **August 2023 – May 2024**
Ultracold Atomic, Molecular, and Optical Lab, William & Mary *Williamsburg, VA*

- Used FEKO and COMSOL software to simulate lens properties of negative index metamaterials for use in a microwave atom trap.

Mixing/Recording Engineer **July 2022 – Present**
Freelance *Nashville, TN*

- Recorded, mixed, and mastered multiple studio albums totaling 170,000+ online streams.
- Perform audio repair such as spectral editing, plosive and sibilance removal, level normalization, and noise reduction for damaged recordings, enabling their use in releases.

Technical Skills

Machine Learning:	PyTorch, Keras/TensorFlow, scikit-learn; CNNs, temporal convolutional networks, time-frequency mask prediction, synthetic training-data generation
Audio & DSP:	Audio feature extraction, STFT/FFT analysis, digital filter and dynamic EQ design, PureData, analog audio processing
Programming:	Python (NumPy, SciPy, pandas, Matplotlib), MATLAB, LaTeX; PyVISA/SCPI instrument automation
Lab & Hardware:	Oscilloscope, spectrum analyzer, VNA, arbitrary waveform generator; analog, digital, and RF electronics; through-hole and surface-mount soldering

Awards

Capstone Prize | *William & Mary Physics Department* **May 2025**
Superior Performance Award | *NSWC: Dahlgren Division* **August 2024**

Publications

- **R. Pamadi**. "Analog Design of an AI-powered Synthesizer." *Honors Thesis, William & Mary*, 2025.